

BRICS INNOVATION THEME • TRACK 1 OFFICIAL SUBMISSION



CivicPulse-BRICS

by Syntax Error

AI for Digital Public Infrastructure (DPI) & Governance: A Sovereign Multilingual AI Platform for Multichannel Citizen Petition Aggregation and CapEx Infrastructure Budget Realignment.

IN India Node (Delhi NCR) • BR Brazil Node (São Paulo) • ZA South Africa Node (Gauteng)

The Problem: Governance & CapEx Friction

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Track 1 Challenge — Fragmented feedback & misaligned capital investments

Fragmented Citizen Feedback

Citizen petitions & development requests live in disconnected departmental channels, voice messages, and paper files with zero unified aggregation engine.

Misaligned CapEx Spending

Millions in capital expenditure (CapEx) are allocated to low-demand projects while critical citizen infrastructure hotspots remain severely neglected.

Unaddressed Infrastructure Failures

Stormwater pipelines, failing power grids, and transit culverts remain unmonitored until catastrophic climate or urban emergencies strike.

Missing Demographic & Risk Data

Lack of cross-analysis combining citizen petitions with population density census data, vulnerability indices, and live IMD meteorological risk vectors.

The Solution: CivicPulse-BRICS Platform

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A Sovereign Multilingual AI Platform & Federated Digital Public Good

01. Multichannel Gateway

Native voice notes, photo defect scans, and an automated WhatsApp & Telegram bot webhook aggregating citizen petitions across diverse languages without requiring app installs.

02. AI Budget Deficit Engine

Cross-analyzes weighted citizen demand against actual government CapEx budgets in real-time, surfacing severe spending deficits and misallocations instantly.

03. Gemini Policy Directives

Generates multi-crore capital budget reallocation directives and strategic national infrastructure megaplans for policymakers and ministers.

Who It Serves: Core Stakeholders

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Empowering citizens, municipal field crews, central ministers, and BRICS nations

1. Citizens & Community Voice

File petitions in native languages via WhatsApp, Telegram, or Web. Voice-to-text translation and real-time status tracking.

2. City Municipal Repair Crews

Automated task queue prioritized by AI severity ratings (1-10) + 1-tap Google Maps turn-by-turn route dispatch for field trucks.

3. Central Ministries & Policymakers

GIS Demand Hotspot Heatmaps, CapEx misalignment alerts, IMD weather risk vectors, and Gemini executive policy directives.

4. BRICS Sovereign Nations

Federated node architecture (India Delhi, Brazil São Paulo, South Africa Gauteng) with cross-border Digital Public Good asset sharing.

AI Approach & Multi-Agent Architecture

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Powered by Google Gemini 2.5 Pro & Flash multi-agent intelligence

Gemini Vision Defect Agent

Analyzes uploaded infrastructure photos, automatically classifying defect type, severity score (1-10), structural hazard rating, and priority code.

Multilingual Speech & Translation

Converts native voice notes in Hindi, Portuguese, or Zulu to text, translates to English, and generates audio synthesis with Google TTS.

CapEx Budget Alignment Engine

Correlates citizen petition density with census demographics, weather risk vectors, and public investment plans to calculate budget misalignment.

Gemini Policy Directive Agent

Synthesizes executive policy recommendations and multi-million dollar CapEx budget reallocation proposals for national ministers.

Live Prototype Demo: Citizen Voice & DPI Gateway

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Multilingual voice recorder, multimodal vision defect scanner & weather HUD

Citizen Gateway Features

- Multimodal AI Vision Scanner: Photo defect classification & 8.7/10 severity rating.
- Multilingual Voice Petitions: Speech recorder with Hindi/English translation toggle.
- WhatsApp & Telegram Webhook: Instant bot petition filing without downloading apps.
- Live IMD Weather Risk HUD: Real-time monsoon and flood risk vectors.



SCREENSHOT DEMO: CITIZEN PORTAL & DPI GATEWAY

URL: <http://localhost:8000/citizen>

Capture & Insert Instructions:

- Voice Grievance Recorder with active wave audio visualizer & Hindi/English toggle.
- AI Multimodal Defect Scanner with uploaded road pothole image & severity score (8.7/10).
- WhatsApp & Telegram bot webhook messaging simulator panel.
- Live IMD Weather Risk HUD with monsoon/rainfall alert banners.

Live Prototype Demo: City Operations & Field Triage

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Real-time repair queue, priority scoring & Google Maps navigation

City Operations Features

- Real-Time Repair Queue: Dynamic ranking by AI severity score (Critical 9.2/10).
- Automated Crew Dispatch: Real-time status tracking for repair trucks (Truck-01 En Route).
- Google Maps Field Routing: 1-Tap turn-by-turn route dispatch for municipal vehicles.
- Operational Efficiency: Zero manual sorting overhead for field engineers.



SCREENSHOT DEMO: CITY OPERATIONS & FIELD TRIAGE

URL: <http://localhost:8000/city-official>

Capture & Insert Instructions:

- Live City Repair Queue with prioritized task cards & CRITICAL (9.2/10) badges.
- Automated Field Crew Dispatch table showing repair truck status (Truck-01 En Route).
- Interactive Google Maps Field Route preview with turn-by-turn navigation overlay.

Live Prototype Demo: National Ministry Command

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GIS demand heatmap, CapEx misalignment matrix & Gemini directives

National Command Features

- Spatial Demand Hotspots: Interactive Leaflet GIS map visualizing high citizen demand zones.
- CapEx Misalignment Matrix: Flags severe budget deficits (88% demand vs 24% budget).
- Gemini Policy Directives: Automated multi-crore capital budget reallocation proposals.
- Public Investment Integration: Connects directly with national capital budgets.



SCREENSHOT DEMO: NATIONAL MINISTRY COMMAND & CAPEX MATRIX

URL: <http://localhost:8000/government>

Capture & Insert Instructions:

- Interactive GIS Spatial Demand Heatmap over Delhi NCR coordinates.
- CapEx Budget Misalignment Matrix comparing citizen demand index (88%) vs allocated budget (24%).
- Google Gemini 2.5 Executive Directives detailing recommended budget shifts.

Live Prototype Demo: BRICS Sovereign Minister Hub

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Multi-nation federation nodes, Gemini megaplans & UN SDG telemetry

BRICS Hub Features

- Multi-Country Jurisdiction: Switch between India (Delhi), Brazil (São Paulo), and South Africa (Gauteng).
- Strategic Megaplans: Automated generation of multi-million infrastructure overhauls (\$45M CapEx).
- UN SDG Telemetry: Live progress tracking against UN SDGs 9, 11, and 16.
- Cross-Border Exchange: Federated DPG data sharing between BRICS nations.



SCREENSHOT DEMO: BRICS SOVEREIGN MINISTER HUB

URL: <http://localhost:8000/central-official>

Capture & Insert Instructions:

- Multi-nation jurisdiction node selector (India Delhi, Brazil São Paulo, South Africa Gauteng).
- Google Gemini 2.5 Strategic Megaplans (\$45M Drainage Infrastructure Overhaul).
- UN SDG Alignment Telemetry (SDG 9, 11, 16 metrics) & DPG Exchange status.

Digital Public Good (DPG) & Open Standards

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Built for open governance, interoperability, and privacy

OpenAPI 3.0 Webhook Specification

Fully compliant standard webhook endpoint (POST /api/v1/dpg/messaging-webhook) for instant integration with national messaging platforms.

Privacy & Security by Design

Anonymized citizen telemetry and salted PBKDF2-HMAC-SHA256 password hashing. Zero persistent tracking of sensitive personal identities.

UN SDG Strategic Alignment

Directly contributes to UN SDG 9 (Industry & Infrastructure), SDG 11 (Sustainable Cities), and SDG 16 (Peace, Justice & Strong Institutions).

Open Data & Interoperability

Exports structured JSON-LD schemas and GIS records compatible with open government portals like data.gov.in and dados.gov.br.

Deployability, BRICS Scale & Google Tech Stack

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Cloud Run containerization, federated jurisdiction nodes & Google Cloud AI architecture

IN India Node (Delhi NCR)

- PM Gati Shakti spatial portal & data.gov.in integration
- Live IMD meteorological weather telemetry HUD
- CapEx alignment with national ministry priorities

BR Brazil Node (São Paulo)

- PAC (Programa de Aceleração do Crescimento) integration
- Brazilian Open Data (dados.gov.br) & Portuguese triage
- Amazonian watershed flood & vulnerability indices

ZA South Africa Node (Gauteng)

- Municipal Infrastructure Grants (MIG) registry integration
- Multilingual Zulu & English speech triage agent
- Township power grid & water resilience tracking

Gemini 2.5 AI

Google Gemini 2.5 Pro & Flash multi-agent defect triage, voice translation, and strategic policy directives.

Google Maps

Interactive spatial pinpointing & 1-tap turn-by-turn routing for municipal repair trucks.

Cloud Run

Serverless containerized microservices running high-performance asynchronous Python ASGI core.

BigQuery GIS

Spatial warehouse (POINT coordinates) for macro analysis + Firebase Cloud Firestore realtime sync.

THANK YOU

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AI for Digital Public Infrastructure (DPI) & Governance • Track 1 Official Submission

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 Live Demo: <http://localhost:8000> •  GitHub: <https://github.com/ParasSingh29/CivicPulse-BRICS> •  Apache 2.0 Open Source DPG